

Program: Diploma in Mechatronics	
Course Code: 6431A	Course Title: Introduction to Mechatronics
Semester: 6	Credits: 4
Course Category: Program Open Elective	
Periods per week: 4(L:4 T: P:0)	Periods per semester: 60

Course Objectives:

- To provide knowledge about the importance and role of automation and mechatronics in manufacturing systems and industries.
- To identify various types of sensors and their applications.
- To identify various mechanical, hydraulic, pneumatic, and electrical actuation systems and their applications
- To gain an understanding of digital communications concepts and to create PLC programmes.

Course Pre-requisites:

Topic	Course code	Course name	Semester
Knowledge of basic Fluid mechanics & Fluid power.		.	
Knowledge of basic Electrical & Electronics			

Course Outcomes:

On completion of the course, the students will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO 1	Understand the importance and need of automation and mechatronics	11	Understanding

CO 2	Describe various types of sensors and their applications.	15	Understanding
CO 3	Explain the various mechanical, hydraulic, pneumatic and electrical actuation systems.	17	Applying
CO 4	Explain the basic PLC architecture and PLC programming concepts	17	Applying

CO – PO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3		2	-	2	-	2
CO2	3	-	-	-	-	-	
CO3	3	-	-		-	-	
CO4	3	2	-	2	-	-	2

3-Strongly mapped 2-Moderately mapped 1-Weakly mapped

Course Outline

Module outcome	Description	Duration (Hours)	Cognitive Level
CO1	Understand the importance and need of automation and mechatronics.		
M 1.01	Understand the importance of automation and the types of automation	2	Understanding
M 1.02	Explain the mechatronic system, its applications, and its elements	4	Understanding
M1.03	Identify simple examples of mechatronic system	4	Understanding
M1.04	Understand the design procedure of a mechatronic system	1	Understanding

Contents:

Automation - definition - need, advantages and disadvantages of automation-Types of automation fixed, programmable and integrated automation-their advantages and disadvantages -basic elements of an automated system- the future challenge of automation
 Mechatronics- Importance of Mechatronics and its applications in modern industries- elements of Mechatronic system- Elements of Measurement System -Control systems and their types- Open and Closed loop control System-basic elements- simple examples of mechatronic control systems- basic washing machine system- automatic camera
 Difference in Traditional and mechatronic design approaches -mechatronics design procedure

CO2	Describe various types of sensors and their applications		
M 2.01	Identify Different types of sensors	1	Understanding
M 2.02	Define the terminology, static and dynamic characteristics of sensors	2	Understanding
M 2.03	Explain the working of different types of proximity, velocity, force, fluid pressure, temperature, light sensors	8	Understanding
M 2.04	Explain the selection of sensors.	1	Understanding
M 2.05	Identify the application of sensors in washing machines, automatic water level controllers and refrigerators.	3	Understanding
	Series Test – I		
Contents: Sensors- different types – Terminology- static and dynamic characteristics, Position & Proximity Sensors-eddy current proximity sensor-optical encoders-pneumatic sensors-proximity switches- Velocity and Motion Sensors-incremental encoder- tachometer generator; Force Sensors; Fluid Pressure Sensors; Flow Sensors; Liquid Level Sensors; Temperature Sensors-Thermistors-Thermocouples- Bimetallic strip-RTD- Light Sensors; Limit switches-principle-types- Selection of Sensors. Application of sensors and controllers in washing machines, automatic water level controllers and refrigerators.			
CO3	Explain the various mechanical, hydraulic, pneumatic and electrical actuation systems.		
M 3.01	Identify hydraulic and pneumatic actuation system	2	Applying
M 3.02	Explain different types of valves and cylinders	9	Understanding
M 3.03	Explain the components of the electrical actuation system	3	Understanding
M 3.04	Explain the principle of working of AC, DC, and stepper Motors	3	Applying
Contents: Hydraulic and Pneumatic System, -layout- Control valves: Directional control valves- spool valve, poppet valve- pilot operated valve -directional valve, Pressure control valves -pressure regulating valve -pressure limiting valve and pressure sequence valves. Flow control valves. Process control valve -Cylinders. - Single acting and double acting -cylinder sequencing-, - diaphragm actuators - rotary actuators. Semi rotary actuators. Electrical actuation systems: mechanical switches, Relays, solid-state switches, diodes, thyristors, triacs, bipolar transistors, and Solenoids. Working principle of types of AC and DC motors and Stepper motors. Mechanical aspects of motor selection.			
CO4	Explain the basic PLC architecture and PLC programming concepts		

M 4.01	Programmable Logic Controller (PLC): Definition; Basic block diagram of PLC.	3	Understanding
M 4.02	Input/Output processing; PLC Programming: Ladder diagram, its logic functions, Latching and Sequencing.	6	Understanding
M 4.03	PLC mnemonics, Timers; Internal relays and Counters; Shift registers; Master and Jump Controls	4	Understanding
M 4.04	.Data handling; Analog input/output; Selection of PLC	2	Applying
M 4.05	Common hardware faults in mechatronic systems- fault detection techniques	2	Understanding
	Series Test – II		

Contents:

Programmable Logic Controller (PLC): Definition; Basic block diagram of PLC; Applications comparison with computer-Input/Output processing; PLC Programming: Ladder diagram, its logic functions, Latching and Sequencing; PLC mnemonics; Timers; Internal relays and Counters; Shift registers; Master and Jump Controls; Data handling; Analog input/output; Selection of PLC.

Factors influencing PLC- Physical, electrical and maintenance

Common hardware faults in mechatronic systems- fault detection techniques for mechatronic components

Text / Reference

T/R	Book Title/Author
T ₁	Mechatronics – W. Bolton, Pearson Education India.
R ₁	A Text Book on Mechatronics – R.K.Rajput, S.Chand& Co, New Delhi.
R ₂	Mechatronics – M.D.Singh & Joshi, Prentice Hall of India.
R ₃	Mechatronics – HMT, Tata McGraw Hill, New Delhi.
R ₄	Mechatronics System – Devadas Shetty, PWS Publishing
R ₅	Exploring Programmable Logic Controllers with applications – Pradeep Kumar Srivatsava, BPB Publications.

Online Resources

Sl.No	Website Link
1	https://youtu.be/zVVITxiec7g
2	https://youtu.be/8Qiy2nf4sp0
3	https://youtu.be/Xl2nWDcy0To
4	https://nptel.ac.in/course.html